The Tree at the Bottom of Everything

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Abstract: Nested similarity—the fact that some things are more alike than others—has a precise mathematical shape: the tree. When we measure distance on a tree by how recently two items shared a common branch, distance obeys a rule that ordinary distance does not: for any three items, the two largest distances are always equal (triadic matching). This rule, and the consequences that flow from it—that every point is its own center, that the scaffold dissolves leaving only relations, and that the tree generates both structure and narrative—appears across physics, biology, linguistics, history, and consciousness. The convergence is not an analogy. It is the same structure appearing at different scales—the simplest one known that can organize nested groupings without contradiction.

Keywords: consilience, convergence, tree structure, triadic matching, cophenetic distance, hierarchical ontology, invariants, relational reality, structural realism, process ontology, generative ontology

1. What We Notice

Look around you. Some things are alike. Some things are different. A cat is more like a lion than it is like a mushroom. A Monday is more like a Tuesday than it is like a birthday. You recognize your friend's face whether they are three feet away or thirty.

This is so ordinary that we almost never stop to ask: what does it mean for two things to be "alike"? And what does this tell us about how the world is put together?

When we say two things are alike, we mean they share something—a feature, a property, a way of behaving. And when we say they are different, we mean they lack that shared something. But the "shared something" is not equally shared. A cat and a lion share being mammals. A cat and a mushroom share only being alive. A cat and a rock share only being made of atoms. So similarity comes in layers—nested layers, like the rings of an onion.

The deeper you go, the more things share. At the outermost layer, everything is different from everything else. At the innermost layer, everything is the same as everything else—because at the deepest level, everything simply *is*.

This nested structure of similarities is not a metaphor. It has a precise shape. And that shape is a tree.

2. The Shape of Grouping

Imagine you are sorting a pile of objects. You put the ones that are most alike into small groups. Then you combine those small groups into larger groups based on looser similarities. Then those into still larger groups. If you draw this process, you get a branching structure—a trunk that splits into big branches, which split into smaller branches, which split into twigs, which end in individual leaves.

This is a tree. Not the kind with bark and photosynthesis—but the mathematical kind: a pattern of branching points where each point represents a grouping decision, and each leaf represents an individual thing.

The tree is the most natural way to represent nested groupings. And it turns out that almost everything we know about the world comes to us already organized this way. The living world is organized as an evolutionary tree—species branching from common ancestors. Languages are organized as family trees—words and grammars branching from shared roots. Ideas are organized as trees of concepts—general ideas branching into specific ones. Even the history of the universe can be drawn as a tree—the initial state branching into galaxies, stars, planets, life.

But here is the surprising thing. When you organize the world as a tree, the way you measure distance between things changes in a way that is completely different from ordinary distance. And this difference has consequences that ripple through everything.

3. How Distance Works on a Tree

In ordinary space, distance adds up in a straight line. If you walk from your house to the store, and then from the store to the park, the total distance you traveled is the sum of the two legs. This is so obvious we never question it. It is how measuring tapes work. It is how maps work.

But on a tree, distance works differently. The distance between two leaves on a tree is not measured by crawling along the branches. It is measured by how far up the tree you have to go to find a branch that they both share—their most recent common branching point.

Think of it this way. You and your cousin share grandparents. You and a stranger share ancestors much further back. The "distance" between you is measured by how recently you shared a branch in the family tree. This is called *tree distance*.

Tree distance follows a rule that ordinary distance does not. The rule is this: for any three things, the two largest distances between them are always equal.

Let that sink in. In ordinary space, if you have three points, the distances between them can be anything—short, medium, long. You can always draw a triangle with three different side lengths. But in a tree-organized world, for any three items, the two largest distances must match. If A and B are very far apart, and B and C are also very far apart, then A and C must be exactly as far apart as whichever of those two is larger. You cannot have three distances all different. The tree forbids it.

This rule is not an opinion. It is a mathematical necessity. Any set of things organized into nested groups, when you measure distance by the level of their most recent common grouping, will obey this rule automatically. It is baked into the structure.

Let us call this rule *triadic matching*: among any three, the two largest distances are equal.

4. Why This Matters: The World Through Tree-Glasses

If the world is organized like a tree, then triadic matching should show up everywhere we look. And it does—in places where we would never have expected it.

When physicists study the fine-grained structure of matter, they find that particles organize into families and subfamilies—a tree of particle types. The “distances” between these particle types, measured by how strongly they interact or how likely they are to transform into each other, obey the triadic matching rule.

When biologists build evolutionary trees from genetic data, the distances between species automatically obey triadic matching—because that is how a tree works. Every genetic distance matrix derived from a real evolutionary tree has this property. If it did not, it would mean the tree was wrong.

When linguists trace how words change over time, they build language trees. The distances between languages, measured by how many word-forms they share, obey triadic matching.

The same pattern appears in the classification of crystals, in the organization of galaxies into clusters and superclusters, in the way human societies split and recombine, in the way memories are stored and retrieved.

Triadic matching is everywhere. Not because nature is “trying” to be tree-like—but because whenever you have nested groupings, you get a tree. And whenever you have a tree, you get triadic matching. It is the signature of hierarchical organization.

A deeper structure: the bridge between the discrete and the continuous. The tree does something else that is remarkable. It connects two worlds that seem completely different: the world of separate, countable things and the world of smooth, flowing change.

Think of the tree of life. Each species is a discrete leaf—a cat is a cat, a dog is a dog. But the process that produced them—evolution—is continuous. Species did not appear in sudden leaps. They branched gradually from common ancestors. The tree captures both: the discrete categories at the leaves (what things *are*) and the continuous history at the branches (how they *became*).

The same structure shows up in music. A musical note is a discrete frequency—an A is 440 Hz. But a melody is a continuous flow from note to note. The musical scale is a tree: the octave is the trunk, the fifth is a major branch, the third is a smaller branch. The tree organizes the discrete notes into a structure that supports continuous movement between them.

This is not just a pattern we impose. It appears to be how nature itself organizes the relationship between the stepwise and the flowing. The tree is where number meets wave, where category meets process. It is the bridge.

5. Every Point Is Its Own Center

Here is another consequence of tree-thinking that is even stranger.

In ordinary space, there is a single center. The center of a circle. The center of a room. The center of the Earth. There is one special point that is equidistant from everything else.

But on a tree, *every point is its own center*. From the perspective of any leaf, the tree looks different. Your leaf sees a world organized around it—with its nearest neighbors being leaves that share a recent branch with it, and distant strangers being leaves that share only the oldest, deepest branch.

Imagine you are a leaf on a vast tree. If you look at the tree from where you sit, it looks like the tree radiates outward from you. Everything is organized relative to your position. But if you move to a different leaf, the tree looks entirely different—organized around that leaf instead. And yet the tree itself has not changed. The branching structure is the same. Only the perspective has shifted.

This is not a metaphor. It is a precise mathematical property of tree structures. In a tree-organized space, there is no privileged center. Every point has an equal claim to being the center of *its own* universe—because from each point, the network of distances radiates outward in a pattern that is structurally identical to the pattern seen from any other point.

This has a direct connection to something cosmologists have observed about our actual universe: no matter which direction we look, the universe looks roughly the same. Every point in space appears to be the "center" of the cosmic expansion. The standard explanation for this is that the universe has no center—it expands equally everywhere. But the tree perspective offers a deeper reason: if the universe is organized hierarchically, then every point genuinely *is* a center, because the structure itself has no privileged location.

6. The Scaffold That Dissolves

Here is where things get subtle.

A tree is a structure. It has branches and branching points and leaves. But what if the tree itself is not "real" in the sense of being a thing you can point to? What if the tree is just a tool—a scaffold that we build to organize our observations, but that dissolves when we look too closely?

This happens in a precise way in physics. When physicists try to calculate how particles behave, they sometimes use a grid of points—a lattice—as a temporary structure to make the calculations possible. The lattice is not supposed to be physically real. It is a computational scaffold. In the final step of the calculation, you let the spacing between lattice points shrink to zero, and the lattice "dissolves," leaving behind only the physical results—numbers that do not depend on the lattice at all.

But here is the twist: the numbers that survive—the *invariants* (quantities that stay the same no matter what measuring system or scaffold you use)—are always comparisons. They are ratios between one thing and another. The ratio of the electron's intrinsic magnetic strength to its charge, for example, is a pure number: approximately $1/137$. This number does not depend on what units you measure with. It does not depend on how fine your lattice was. It is a ratio of ratios—a comparison of comparisons—that survives the dissolving of every scaffold.

This points to something deep. The "scaffold"—whether it is a lattice, a tree, a coordinate system, or a choice of units—is not part of reality. It is a tool we use to describe reality. When we let the scaffold dissolve, what remains are the invariants: dimensionless numbers, comparisons, relations. These are the things that are "really there."

And what are these invariants? They are relations. They are not things but comparisons between things. The fundamental reality, when you strip away every scaffold, every coordinate system, every choice of units, every arbitrary convention, is purely relational.

A note of caution: patterns and coincidences. When you search for patterns among hundreds of numbers, you will inevitably find near-matches that look meaningful. A scan of 600 pairs of physical constants, tested against 9 targets, would be expected to produce about 100 coincidences at the 1% tolerance level purely by chance. The handful of near-matches that appear are not evidence of hidden structure—they are what randomness looks like. The genuine discovery is the pattern that survives: the invariants, the triadic matching, the tree. These are not statistical flukes. They are structural necessities.

7. The Generosity of the Tree

If this sounds abstract, consider how it plays out in domains far from physics.

Language

Every time you speak a sentence, you are navigating a tree. The noun is the "thing" you are talking about—a leaf on the tree of concepts. The verb is the action or relation—the branch you traverse to get from one leaf to another. Nouns are the nodes. Verbs are the paths between them.

This is not a linguistic theory; it is a direct consequence of how tree structures encode information. You cannot describe anything without picking a leaf (what you are talking about) and a path (what is happening to it or because of it). The noun/verb distinction—found in every human language—is not an accident of grammar. It is a structural necessity of tree-organized communication.

Consider what happens when you tell a story. You begin at a leaf (a character, a situation). You trace paths to other leaves (actions, consequences, other characters). The story is the traversal of the tree. And because every point is its own center, every story has a narrator—a leaf from whose perspective the traversal is organized. Change narrators, and the same events become a different story. The tree has not changed. Only the center has shifted.

History

The past is not a single line. It is a tree of possibilities. At every moment, many futures were possible. Only one was realized. But the unrealized branches are not nothing—they are part of the structure that gives meaning to what actually happened. When we say “things could have been different,” we are invoking the branches that were not taken.

History is the narrative of which branches were selected and which were pruned. And because every point on the tree is its own center, every generation rewrites history from its own perspective—selecting different branches as significant, different ancestors as worthy, different events as pivotal. History is not “what happened.” It is the tree from which “what happened” is selected.

This explains why historical disputes are rarely about facts and almost always about significance. Two historians can agree on every event and still produce radically different histories—because they are centering the tree at different leaves.

Consciousness

You are not a single thing. You are a tree of selves. The you of this morning is a close neighbor to the you of last night—they share a recent branch. The you of ten years ago is a distant relative—sharing only a deep branch. And yet there is continuity: a path through the tree that connects all these selves.

Consciousness is not a static substance but the ongoing process of navigating this tree—of being a leaf that is constantly growing new branches while remaining connected to its past. Every moment of awareness is a branching decision: which thought to follow, which memory to retrieve, which action to take. The self is not the leaf but the path—the accumulated history of branches taken.

This is why a recovering addict can challenge a doctor’s authority: the addict has navigated branches of experience that the doctor, however educated, has not. The tree of lived experience is different from the tree of book knowledge. And because every point is its own center, there is no privileged perspective from which one tree is “more correct” than the other. This is not relativism. It is structuralism. The tree admits multiple centers, and each center yields a valid description of the whole.

The Ladder of Understanding

There is a recurring pattern in how people come to know things. At the bottom of the ladder is raw experience—data, sensation, observation. One rung up is the selection of what to pay attention to. Another rung is the naming of patterns. Another is the drawing of conclusions. Another is the formation of beliefs. Another is the taking of action based on those beliefs.

This ladder is a tree. At each rung, many branches are possible. The data you select determines the patterns you see; the patterns you name determine the conclusions you draw; the conclusions you draw determine the beliefs you form; the beliefs you form determine the actions you take. And because every point is its own center, two people can start from the same raw data and arrive at entirely different actions—not because either is wrong, but because they climbed different branches of the same tree.

Understanding this does not mean all conclusions are equal. It means that to understand why someone believes what they believe, you must trace their path up the tree—not just argue with the leaf they are standing on.

8. What It All Means

We started with a simple observation: things are grouped. Some things are more alike than others. This grouping has a shape—the shape of a tree.

From this single observation, a cascade of consequences follows:

- **Distance on a tree is not ordinary distance.** It obeys triadic matching: for any three things, the two largest distances are equal. This signature appears wherever hierarchical organization is present—in physics, biology, linguistics, and beyond.
- **Every point is its own center.** Because the tree has no privileged leaf, every perspective is equally valid as a center from which to view the whole. This explains why the universe looks the same in every direction, why every generation rewrites history, and why no single viewpoint can claim absolute authority.
- **The scaffold dissolves, leaving only relations.** The tree itself—the branches, the branching points, the particular organization—is not “real” in a fundamental sense. It is a tool. What survives when the tool is removed are the invariants: pure numbers, comparisons, relations. These are not things but comparisons between things. The fundamental reality is relational.
- **The tree bridges the discrete and the continuous.** The same tree that organizes stepwise groupings—species into genera, words into families—also mediates the relationship between the countable and the smooth, between the particular and the universal. The tree is where number meets wave.
- **The tree generates both structure and narrative.** A tree is not static. It grows. New branches form. Old branches are pruned. The tree is a generator—it constantly produces new leaves and new groupings. What we call “laws of nature” are the

invariants that constrain this generation. What we call “history” is the record of which branches were taken.

The convergence across domains is not a coincidence. It is not an analogy. It is the same structure appearing at different scales and in different media—and, as far as we can tell, it is the simplest structure capable of organizing nested groupings without contradiction.

Why This Matters for You

You are not a passive observer of a tree-structured world. You are a leaf on it, navigating it every moment of every day. Every choice you make—what to pay attention to, what to say, what to do next—is a branching. You trace a path through a tree that no one else has ever traced, from a center that no one else occupies. And at the end of your life, the tree you have grown—your relationships, your memories, your decisions—is the only one of its kind in the history of the universe.

This is not poetry. It is structure. The tree is not a metaphor for your life; your life is an instance of the tree. Every choice is a branching. Every relationship is a shared branch-point. Every memory is a leaf. And every moment of awareness is the realization that you are both a leaf on the tree and a center from which the entire tree can be seen.

The world is not made of things. It is made of relations. And the shape of those relations—the deep geometry that underlies everything from particles to poems, from galaxies to grammar, from the first moment of the universe to the next thought in your mind—is the shape of a tree.